

BFP - THEORETICAL MODALITY - EXTRAORDINARY CALL 2026

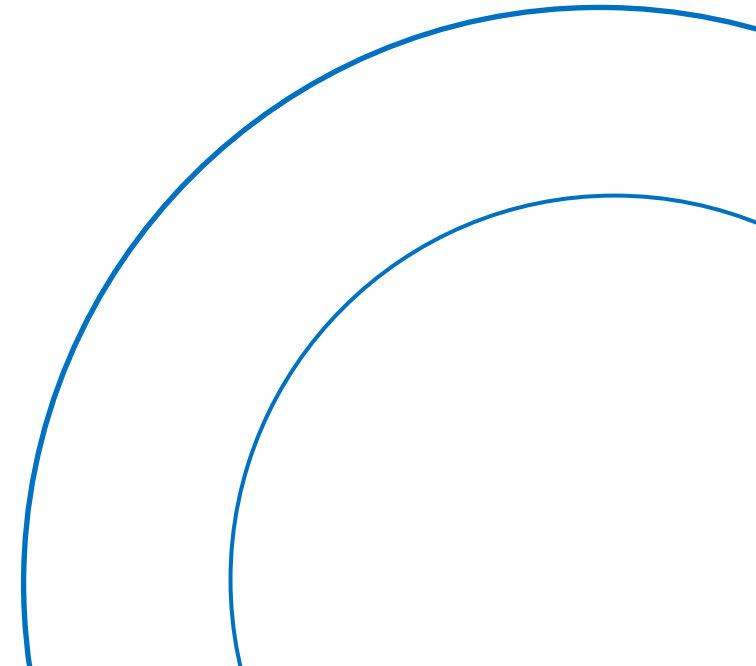
Beyond the Health Bar:

Analysis of Recurring Patterns in the Design of Boss Encounters in Videogames

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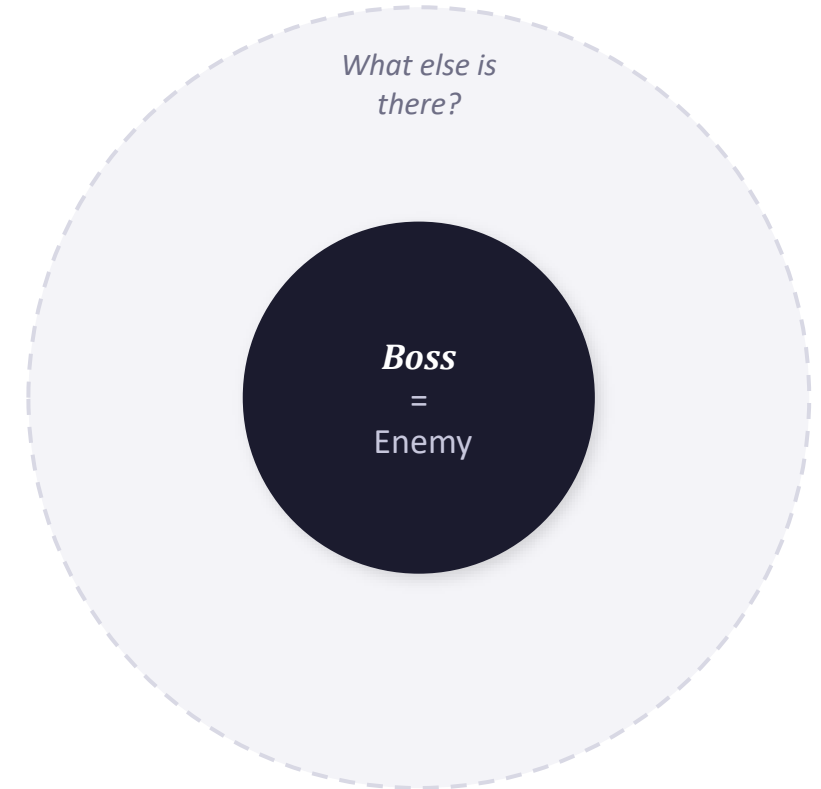
Interactive Product Design

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Bosses Are Still Treated as “Bigger Enemies”

- Academic literature limits the term “**boss**” to a subcategory of **enemy design**: the “**boss monster**.”
- This **focuses** the design **on combat and mechanics**, narrowing down creativity to the mechanical design; **leaving out narrative or spatial** confrontations.
- So... Is there really **no boss** without an **enemy**?

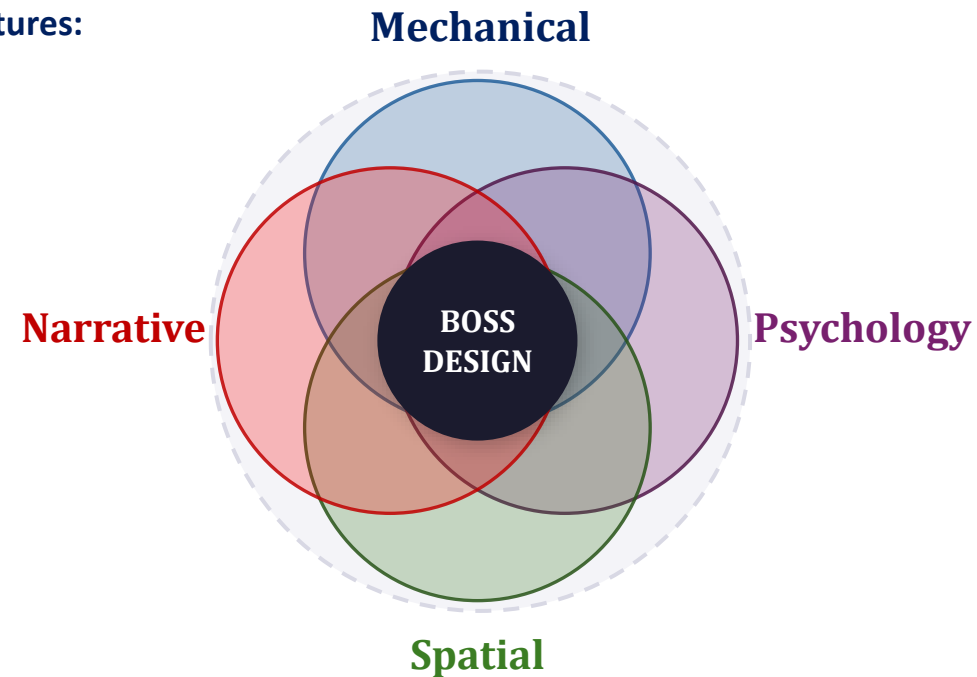


1. HYPOTHESIS & OBJECTIVES

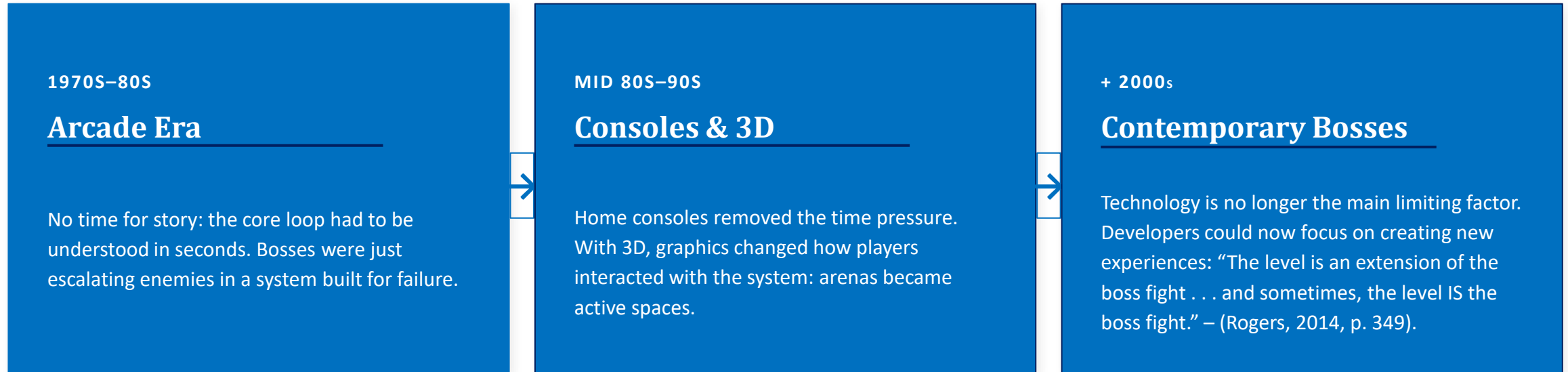
Redefining the Boss Encounter

Boss encounters, regardless of genre, share recurring patterns across design dimensions that can be abstracted into a single design framework.

Bosses as complex multidimensional structures:



From Mechanical Obstacle to Holistic Experience



Technology and Industry progressively evolved, leading to a shift in the design mentality, allowing for more complex experiences.

2. STATE OF THE ART

Four Dimensions of the Design

1 Mechanical

The boss as the **culminating challenge of a learning arc**; **patterns** must be calibrated to keep the player in a **flow state**.

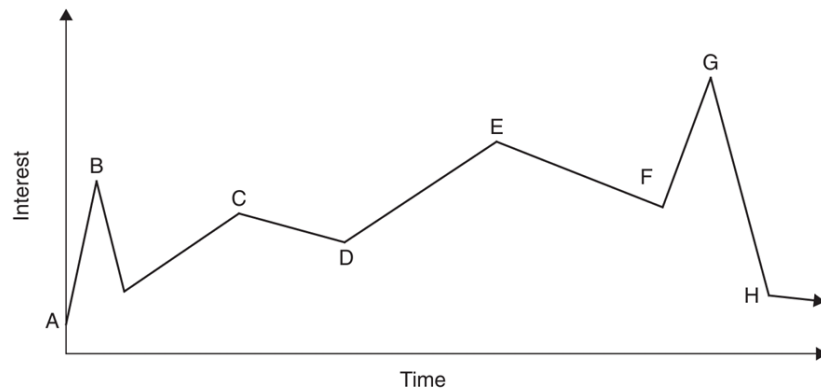


Figure 2. Graphic interpretation of the Interest Curve. Source: (Schell, 2008, p. 248)

2 Narrative

The boss as **crises** and **climax** of the **Three-Act Structure**. It is the medium that allows the plot to unravel.

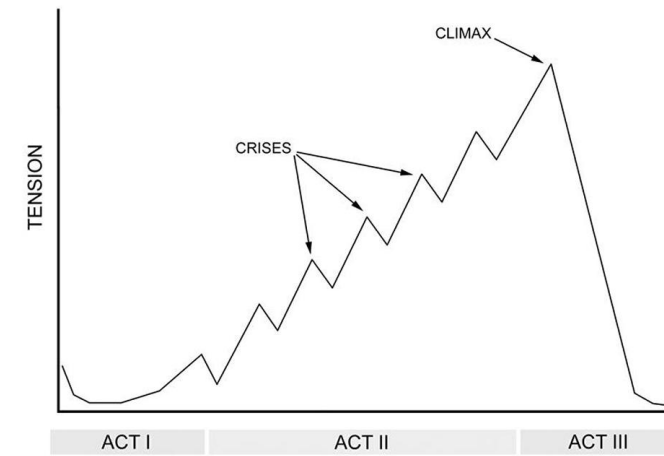


Figure 4. Representation of the Tension Map within the Three-Act Structure. Source: (Schell, 2008, p. 32).

2. STATE OF THE ART

Four Dimensions of the Design

3 Spatial

Space as an **active part of the challenge**: it communicates, evolves, and directs the player's attention.

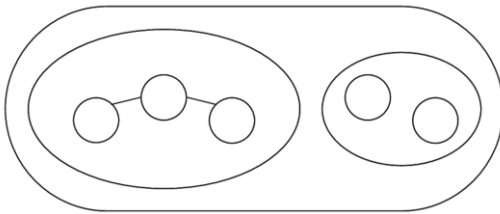


Figure 5. Visual example of a Nested Space. Source: (Schell, 2010, p. 134).

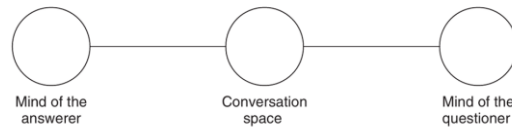


Figure 6. Visual mockup of the space created within the information exchange between two players in a Zero dimensional space. Source: (Schell, 2010, p. 135).

4 Psychological

The **failure-improvement cycle** turns obstacles into validators of the player's progress.



Figure 7. Visual representation of the Failure-Improvement Cycle. Source: (Juul, 2013, p. 60)

Testing the Hypothesis Against Three Genres

MECHANICAL AXIS

Monster Hunter Wilds

Combat-centered design. A traditional boss, used as the control case.

NARRATIVE AXIS

Dispatch

No combat system, no enemies: narrative-centered game. The most direct test of the hypothesis.

SPATIAL AXIS

Portal

No traditional combat: challenge is purely spatial.

Same analytic table for all three

Every encounter is examined under the same four dimensions: mechanical, narrative, spatial, and psychological; through a **qualitative**, pattern-based comparison.

4. DEVELOPMENT

Testing the Hypothesis Against Three Genres



Five Principles of Boss Design

I **Convergence of Curves** Narrative, difficulty, and learning curves are **one cyclical pattern** seen from three angles: **Hook, Build Up, and Resolution**

II **Communicating the Challenge** Telegraphing turns **difficulty into fairness**: it enables **the failure-improvement cycle**.

III **Space as an Active Element of the Challenge** The level is **part of the challenge itself**, not a decorative element.

IV **Ludonarrative Coherence** Every element of the encounter must communicate **coherent information**.

V **Emotional Validation** The **emotional payoff** is **proportional** to what the player **invested** to reach it.

I

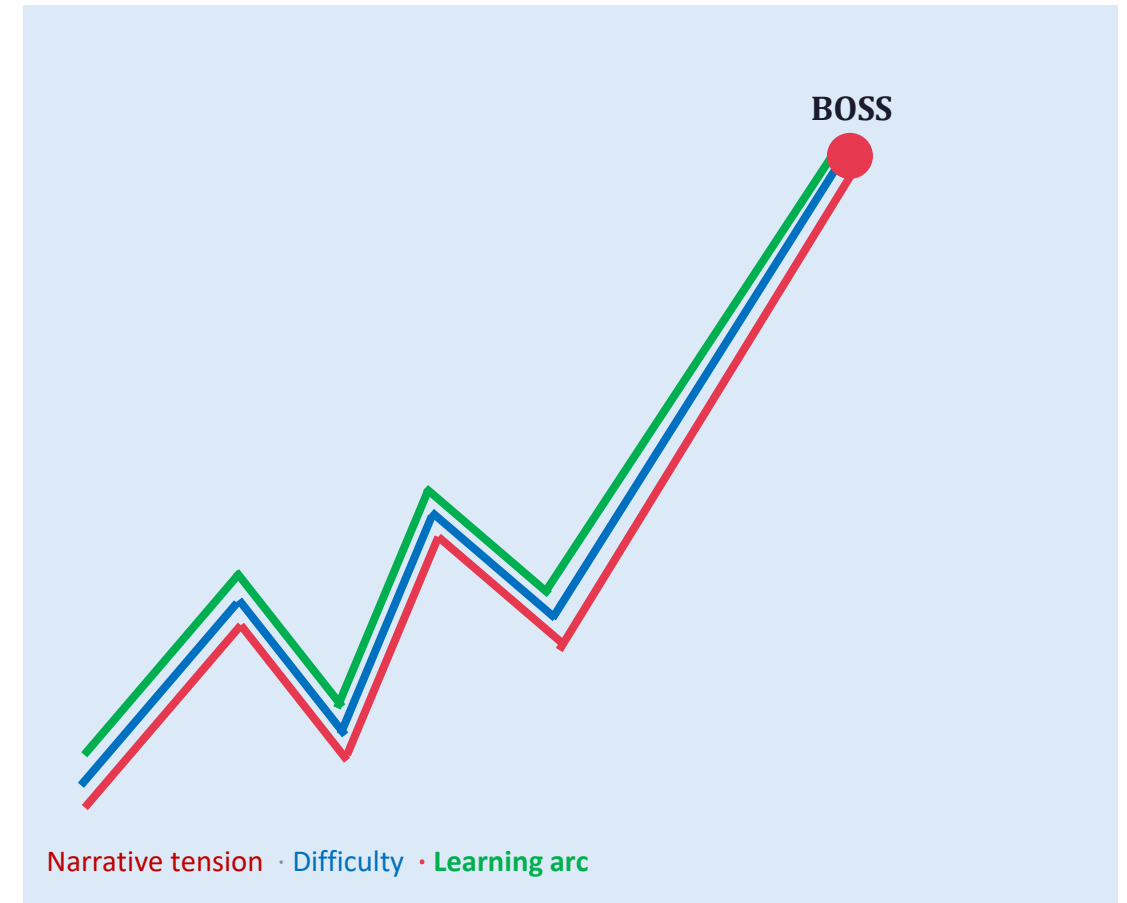
CONVERGENCE OF CURVES

Skolnick's three-act structure, Schell's interest curve, and Rogers' ramping gameplay describe the same cyclical pattern from three different perspectives:

- **Narrative arc (Skolnick):** three acts, escalating crises, climax
- **Interest curve (Schell):** hook, escalation, climax
- **Ramping gameplay (Rogers):** introduce, master, intensify

Same pattern, at every scale:

Act, Level, Individual Pattern → Each one peaking higher until the boss.



This is merely a visual graph; it does not represent the real values of the curves.

II

COMMUNICATING THE CHALLENGE

Telegraphing is what makes a **challenge feel fair** rather than arbitrary. It enables Juul's failure-improvement cycle: the player **fails, understands** the cause, and **improves**.

A difficult game punishes arbitrarily.

A challenging game can be read and overcome.

III

SPACE AS AN ACTIVE ELEMENT OF THE CHALLENGE

The level is not decoration: it is a **part of the challenge**. It **communicates** narrative, **evolves** with the encounter, and **guides** the player's attention.

A decorative level only frames the encounter.

An active level takes part in it.

IV LUDONARRATIVE COHERENCE

No channel communicates in isolation. Visual design, mechanical behavior, narrative, and space must all convey the **same meaning and information**.

*When coherence is achieved,
The player is never told what's happening: the design already shows it.*

V EMOTIONAL VALIDATION

Juul frames play as an emotional gamble: the **more effort invested**, the **greater** the **payoff** at resolution. This principle depends on the correct implementation of the previous four, as it is transversal to them.

Unlike a film, the player's own agency, repeated failure, learning, and mastery, is what **makes victory feel earned**.

SO, WHAT IS A BOSS?

Not the enemy with the most health...

The point where mechanics, narrative, space, and psychology converge into a single moment of maximum tension.

Limits of the Study

1

GENRE SCOPE

Only three, deliberately distinct genres were analyzed. A **broader sample** would help the study be **more precise**.

2

EXTREME CASES UNTESTED

“**Bossless**” or highly unconventional games **were not included**, which would test the framework at its limits.

3

AN OPEN QUESTION

Does every videogame have a boss?

If this framework holds, yes. In whatever form the game's design takes.

Where This Goes Next

1

Genre-specific frameworks

Research genre-specific frameworks that will allow for more **precise, specialized versions** of this framework for individual genres.

2

Competitive multiplayer

Test whether the principles established in this framework hold up in **competitive multiplayer games**.

3

Practical application

Design an original boss encounter from the framework and evaluate the resulting experience.

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Thank you.